

Title: Critical Minerals, Critical Pressures: Monetary Transmission in a Mineral-Scarce Euro Area

Applicant: Yagmur Menzilcioglu

Topic Area: The transmission of monetary policy through the financial system

MOTIVATION

The euro area’s clean energy and digital transitions are rapidly increasing dependence on critical minerals such as lithium, nickel, cobalt, and copper. These minerals are essential inputs into low carbon energy technologies, such as, batteries and electrical equipment, digital infrastructure, and defense systems, yet are sourced almost entirely from outside the euro area. Because supply is geographically concentrated and slow to adjust, the region is increasingly exposed to persistent upstream price shocks. Recent episodes, such as the lithium price surge of 2021–2022, illustrate how mineral markets can generate large, long-lasting price swings driven by capacity constraints and synchronized global demand.

For the euro area, mineral price shocks propagate through both production networks and financial markets. On the real side, mineral-intensive sectors—including autos, machinery, and electrical equipment—experience rising costs and delayed investment. Through input-output linkages, these shocks can generate cost-push inflation and sectoral inflation heterogeneity. On the financial side, firms exposed to mineral inputs or clean technology demand face volatile valuations and uneven funding conditions. These changes propagate differently through banks, investment funds, and other non-bank financial intermediaries (NBFIs) that together mediate credit in the euro area.

These channels interact directly with the ECB’s monetary transmission mechanism. When inflation originates from mineral-driven supply pressures, the sensitivity of inflation and investment to policy rate changes may weaken. At the same time, the ECB’s balance sheet tools—such as collateral frameworks, corporate bond holdings, and targeted refinancing operations—may gain influence over financing conditions in mineral-exposed sectors. Despite these emerging pressures, mineral-driven shocks remain absent from mainstream research on euro area monetary transmission, which has focused primarily on oil prices, demand shocks, or financial disturbances.

This project develops the first integrated empirical–theoretical framework for understanding how critical mineral shocks shape the strength, speed, and heterogeneity of monetary transmission in the euro area, with explicit attention to the roles of banks, NBFIs, and ECB balance sheet policies.

RESEARCH QUESTIONS

Main question. How do global critical mineral price shocks shape the strength, speed, and heterogeneity of monetary policy transmission in the euro area, given its production structure, financial system, and reliance on market-based finance?

Operational sub-questions: (i) **Real–financial propagation:** How do mineral shocks feed into euro area inflation, sectoral investment, and financial conditions, and through which mechanisms do banks and NBFIs amplify or absorb these effects? (ii) **Monetary and balance sheet transmission:** How does the transmission of ECB policy rates and balance sheet tools change when inflation originates from mineral-driven supply pressures rather than demand factors?

RELATION TO EXISTING LITERATURE

The project builds on three strands of work. (i) **Commodity and mineral markets:** Javorcik and Schweiger (2024) and Miranda-Pinto (2024) emphasize low supply elasticity, concentrated extraction, and persistent price cycles in critical minerals; (ii) **Euro area monetary transmission:** Evidence from Acharya et al. (2019) and Capiello et al. (2021) highlights how monetary policy transmission in the euro area is shaped by bank balance-sheet strength, sovereign-bank linkages, and the rising influence of NBFIs that respond more to long-term rates and asset-market conditions; (iii) **Production networks:** Research by Acemoglu, Ozdaglar, and Tahbaz-Salehi (2015) shows that upstream sectoral shocks propagate through input-output networks and multi-tier supply chains, amplifying their effects on inflation, output, and

investment across the broader economy. No existing research integrates mineral-driven upstream shocks, sector-financial heterogeneity, and ECB transmission channels. This project fills that gap.

EMPIRICAL STRATEGY

(1) Sectoral mineral intensity and pass-through. Using Eurostat input-output tables and sectoral price data, I will compute mineral intensity of sectoral value added and investment. Pass-through regressions will quantify how mineral shocks propagate into producer and consumer inflation across countries and sectors, identifying where cost-push pressures and inflation persistence are most pronounced.

(2) Financial exposures of banks and NBFIs. I will combine publicly available ECB statistics—including Balance Sheet Items (BSI), investment-fund data, securities issues and holdings by sector (SHSS), and aggregated data with commercial firm-level datasets (e.g., Orbis, Refinitiv, Bloomberg) to map the exposure of banks and NBFIs to mineral-intensive and clean technology firms across sectors and countries. This will allow the construction of sector-level credit and portfolio exposure indicators using publicly accessible holdings and issuance data, complemented by commercial datasets to classify firm-level mineral intensity. While security-level holdings data such as SHS and AnaCredit are not accessible under the fellowship, the proposed strategy remains feasible and can be scaled further through future collaboration with data-access partners or central bank networks.

(3) Interaction of mineral shocks with monetary surprises. Using high-frequency identification of ECB policy surprises, I will estimate how asset prices, credit spreads, and lending conditions respond when mineral shocks overlap with monetary tightening or easing. These elasticities will discipline the calibration of the theoretical model’s transmission parameters.

MODEL FRAMEWORK, CONTRIBUTIONS, AND ECB RELEVANCE

I will develop a multi-sector euro area New Keynesian framework that embeds: (i) mineral-intensive upstream inputs with slow supply adjustment; (ii) sectoral heterogeneity in mineral use and pass-through; and (iii) banks and NBFIs subject to balance sheet frictions, with ECB policy rate and balance sheet tools directly shaping financial conditions. Disciplined by the empirical results, the model will quantify how mineral-driven shocks propagate through production networks and financial intermediaries, when policy rate transmission weakens, and how balance sheet instruments affect financing conditions in mineral-exposed sectors.

The project delivers three core outputs: (i) new exposure measures for mineral- and finance-driven vulnerabilities across euro area sectors and intermediaries; (ii) reduced-form evidence on how mineral shocks interact with monetary policy surprises; and (iii) a calibrated model generating policy counterfactuals for rates and balance sheet tools under mineral-driven inflation. These findings directly inform the ECB’s assessment of monetary transmission in an economy where market-based finance is expanding and supply-side pressures increasingly shape inflation dynamics.

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